DATA CENTER SIMULATION PERFORMANCE REPORT

Workload Profile: NORMAL_DAY | Timestamp: 2026-08-23 19:49:28

OVERALL SLA VERDICT: FAIL (3/4 Checks Satisfied)

1. Experiment Summary & Identification

Parameter	Value
Profile / Workload	normal_day
Execution Duration	600.0 s
Start Time (T1)	1787494167.0858684
End Time (T2)	1787494767.0858684
Total Requests Executed	8275
Primary Bottleneck	None
Statistical Confidence	HIGH
Failure Classification	SLA_NON_COMPLIANCE, HTTP_REQUEST_FAILURES

2. SLA Scorecard & Compliance Targets

SLA Metric	Measured Value	Target Limit	Result
P95 Response Time	110.0 ms	500.0 ms	PASS
Error Rate	1.0 %	1.0 %	PASS
Peak CPU Utilization	82.37 %	80.0 %	FAIL
Peak Memory Utilization	18.95 %	90.0 %	PASS

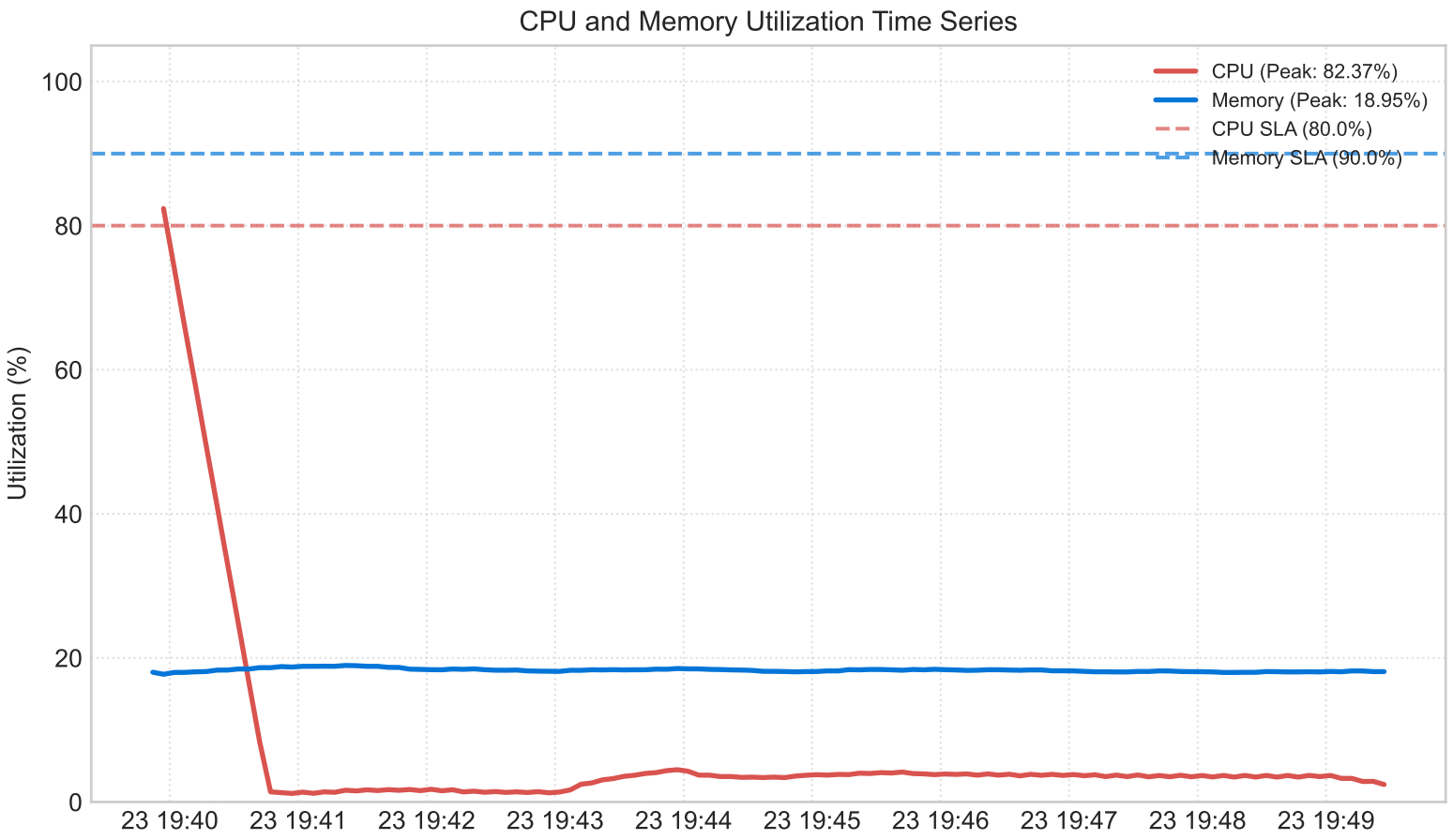
3. Executive Conclusion & Synthesis

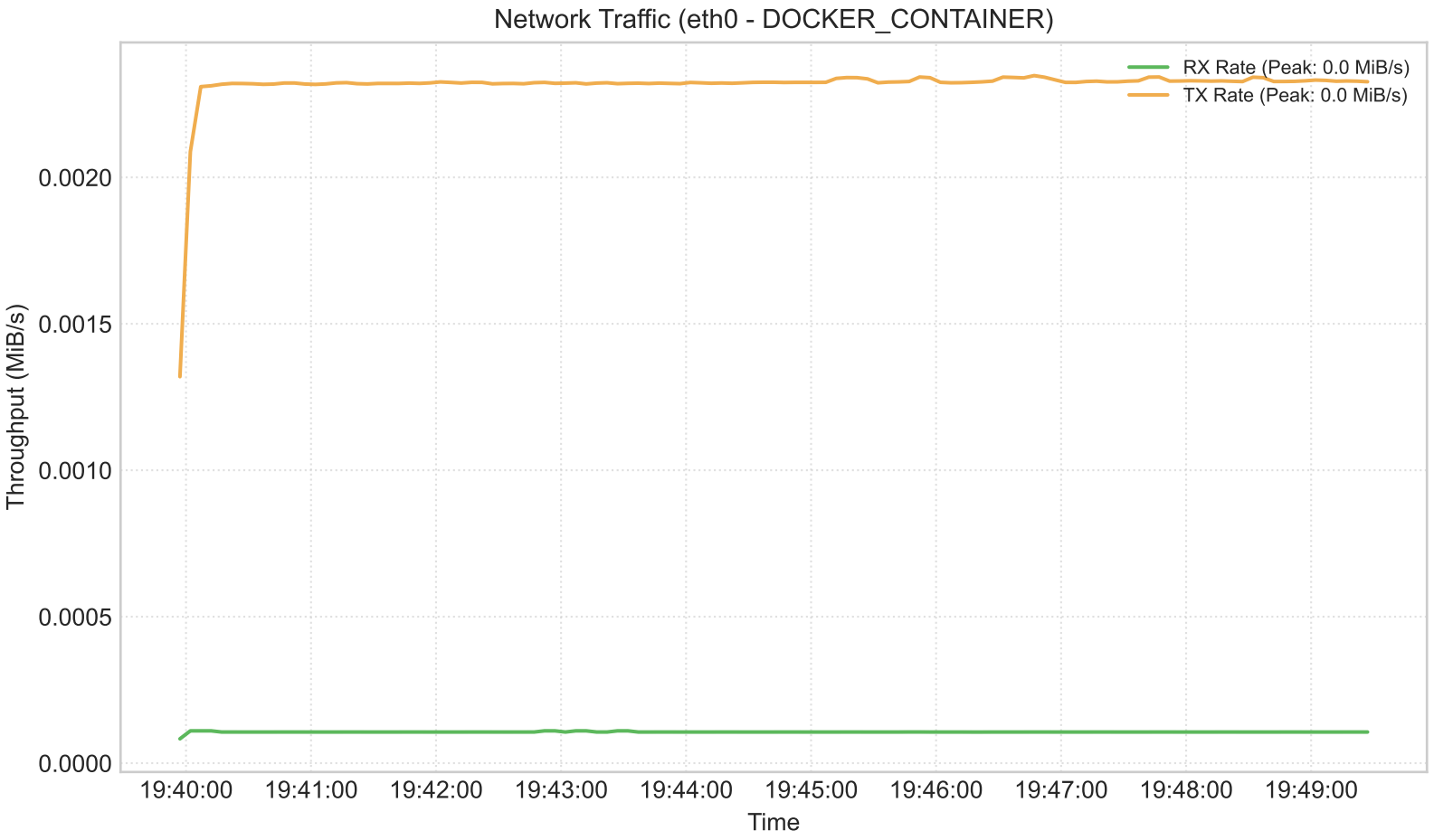
The experiment executed 8275 requests with overall SLA status 'FAIL' (3/4 checks satisfied). No container crashes or service interruptions were detected by the watchdog. Primary bottleneck assessed as 'None' (Confidence: HIGH).

4. Diagnostic Observations & Failure Evidence

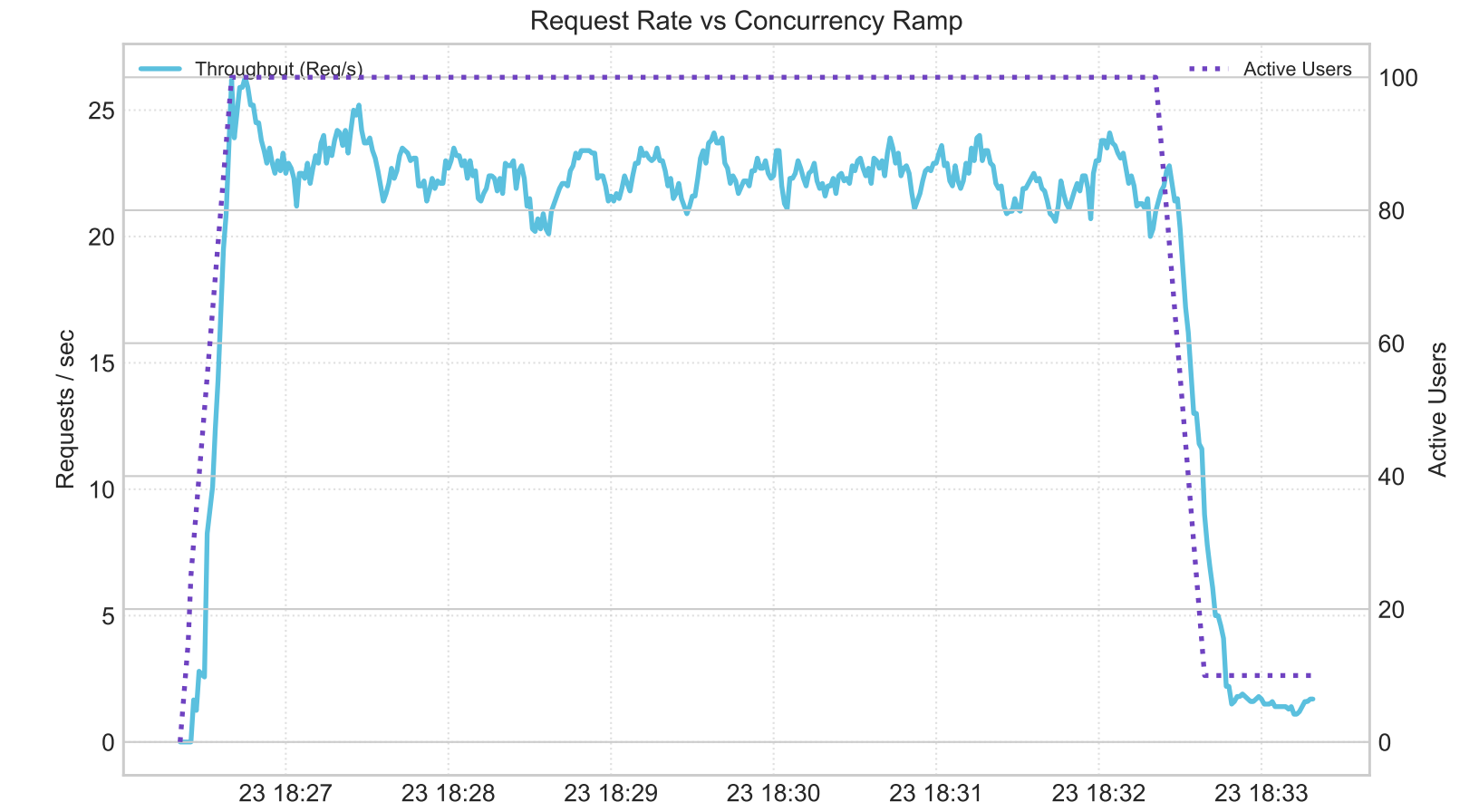
• CPU SLA breach: peak 82.37% exceeded threshold 80.0%.
• All measured system and load parameters remained within safe thresholds.
• Locust recorded 83 failed requests (1.0% error rate).
• Failure on POST /login: HTTPError('429 Client Error: Too Many Requests for url: /login') (80 occurrences)
• Failure on GET /health: RemoteDisconnected('Remote end closed connection without response') (2 occurrences)

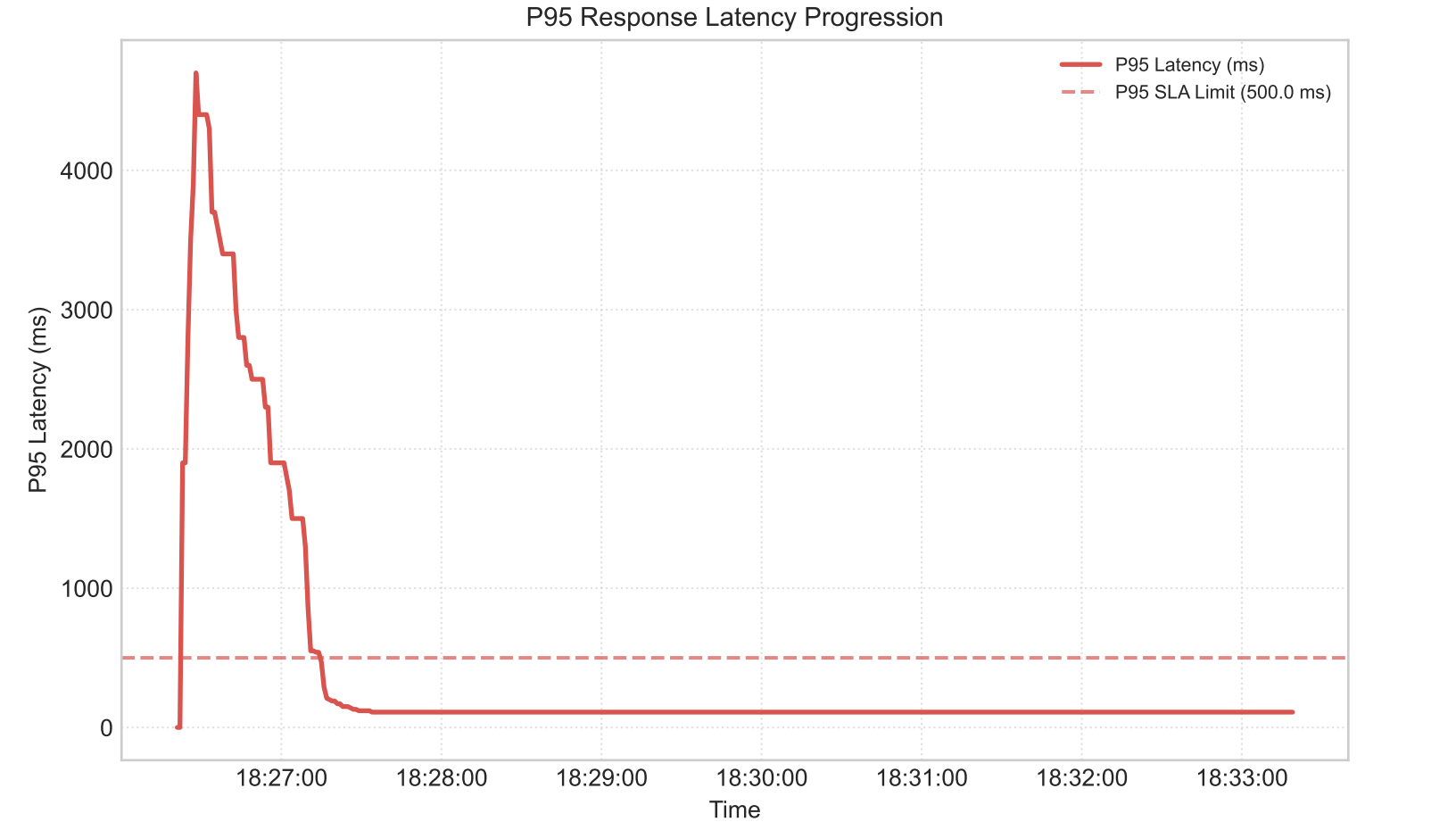
System & Hardware Resource Dynamics





Locust Workload & Latency Profile





Endpoint Breakdown & Chronological Event Timeline

1. HTTP Endpoint Performance Breakdown

Endpoint	Requests	Failures	Avg Latency	P95 Latency	RPS
GET /api/data	2621	0	75.61 ms	120.0 ms	6.24
GET /health	3483	2	13.64 ms	31.0 ms	8.29
POST /login	100	80	1106.68 ms	4400.0 ms	0.24
POST /logout	19	0	29.79 ms	190.0 ms	0.05
GET /me	1664	1	21.51 ms	33.0 ms	3.96
POST /profile/update	388	0	91.15 ms	110.0 ms	0.92

2. Chronological Event Timeline

Time	Event Source & Type	Category	Severity	Description
18:26:21	[LOCUST] LOAD_STARTED	OBSERVED	INFO	Workload generation began with 0 users
18:26:40	[LOCUST] LOCUST_PEAK_USERS	OBSERVED	INFO	Locust concurrency reached peak of 100 active users
18:26:45	[LOCUST] LOCUST_PEAK_RPS	OBSERVED	INFO	Locust throughput reached peak of 26.3 req/s
19:39:27	[RUNNER] EXPERIMENT_START	OBSERVED	INFO	Experiment started at T1 (1787494167.0858684)
[PROMETHEUS]	PROMETHEUS_CPU_PEAK	OBSERVED	WARNING	CPU utilization peaked at 82.37%
[SLA_ENGINE]	SLA_BREACH_DETECTED	INFERRED	CRITICAL	SLA Target Breached: Peak CPU Utilization was 82.37 % (...
[PROMETHEUS]	PROMETHEUS_MEMORY_PEAK	OBSERVED	INFO	Memory utilization peaked at 18.95%
[PROMETHEUS]	PROMETHEUS_NETWORK_PEAK_RX	OBSERVED	INFO	Network receive rate peaked at 0.0 MiB/s on eth0
[PROMETHEUS]	PROMETHEUS_NETWORK_PEAK_TX	OBSERVED	INFO	Network transmit rate peaked at 0.0 MiB/s on eth0
19:49:27	[RUNNER] EXPERIMENT_END	OBSERVED	INFO	Experiment completed at T2 (1787494767.0858684)